

Corollary 3.8 (Variance of $U_n(h)$). *Granted Assumption 2.1 (p, q, ρ) is satisfied. Then there is a universal constant $C' \in \mathbb{R}_+$, such that*

$$\mathbb{E}\left[\left(\frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} h(X_i, X_j) - \theta\right)^2\right] \leq C' n^{-1}.$$

Proof.

$$\begin{aligned} & \mathbb{E}\left[\left(\frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} h(X_i, X_j) - \theta\right)^2\right] \\ & \leq 3 \left(\frac{2}{n(n-1)}\right)^2 \left\{ \mathbb{E}\left[\left(\sum_{1 \leq i < j \leq n} h_2(X_i, X_j)\right)^2\right] + \frac{(n-1)^2}{2} \mathbb{E}\left[\left(\sum_{1 \leq i \leq n} h_1(X_i)\right)^2\right] \right\}. \end{aligned} \quad (3.44)$$

The first term inside the curly parentheses of (3.44) is of order n^2 by Corollary 2.6. The second term inside the curly parentheses is of order $n^2 \cdot n$. Indeed, let X' and X'' be two independent random variables with law $\mathbb{P}_{X_0}$, then by Assumption 2.1 (p, q, ρ)

$$\begin{aligned} \sum_{1 \leq i \leq j \leq n} \mathbb{E}[h_1(X_i)h_1(X_j)] &= \sum_{1 \leq i, j \leq n} \mathbb{E}[(h(X_0, X') - \theta)(h(X_{j-i}, X'') - \theta)] \\ &= \sum_{1 \leq i, j \leq n} \mathbb{E}[(h(X_0, X') - \theta)(h(X_{j-i}, X'') - h(X_{j-i}^{(j-i)}, X''))] \\ &\leq \|h(X_0, X')\|_q \sum_{1 \leq i, j \leq n} L \|d(X_{j-i}, X_{j-i}^{(j-i)})^\rho\|_p \lesssim n. \end{aligned}$$

This completes the proof. $\square$

3.4. Auxiliary results

The bottleneck distance between persistence diagrams D_1, D_2 is given by

$$W_B(D_1, D_2) = W_\infty(D_1, D_2) = \inf_{\gamma} \sup_{p \in D_1} \|p - \gamma(p)\|_\infty$$

where γ is the set of bijections between the multi-sets D_1 and D_2 (a point with multiplicity $m > 1$ is considered as m disjoint copies) and $\|p - q\|_\infty = \max\{|x_p - x_q|, |y_p - y_q|\}$.

Proposition 3.9. *[Chazal et al. (2014) [9], Cohen-Steiner et al. 2007 [10]] Let X, Y be two totally bounded metric spaces. Let $\text{PD}(X)$, resp., $\text{PD}(Y)$ be the persistence diagrams obtained from X , resp., Y using the Čech or Vietoris-Rips filtration. Then*

$$W_B(\text{PD}(X), \text{PD}(Y)) \leq 2d_{GH}(X, Y)$$

where d_{GH} is the Gromov-Hausdorff distance.

Lemma 3.10. *Let X, Y be two finite subsets in a metric space (M, d) . Let $\text{PD}(X)$, resp., $\text{PD}(Y)$ be the persistence diagrams obtained from X , resp., Y using the Čech or Vietoris-Rips filtration. Then*

$$W_B(\text{PD}(X), \text{PD}(Y)) \leq 2d_H(X, Y)$$

where

$$d_H(X, Y) = \max \left\{ \sup_{x \in X} \inf_{y \in Y} d(x, y), \sup_{y \in Y} \inf_{x \in X} d(x, y) \right\}$$

is the Hausdorff distance between X and Y .

Proof. We apply Proposition 3.9 and use $d_{GH}(X, Y) \leq d_H(X, Y)$ in the present case because the identity on M is an isometric embedding. $\square$

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